

**❑ Significance and Determination of
BOD & COD**

❑ E Waste

❑ Green Chemistry

BIOLOGICAL OXYGEN DEMAND (BOD)

WHAT IS BOD ?

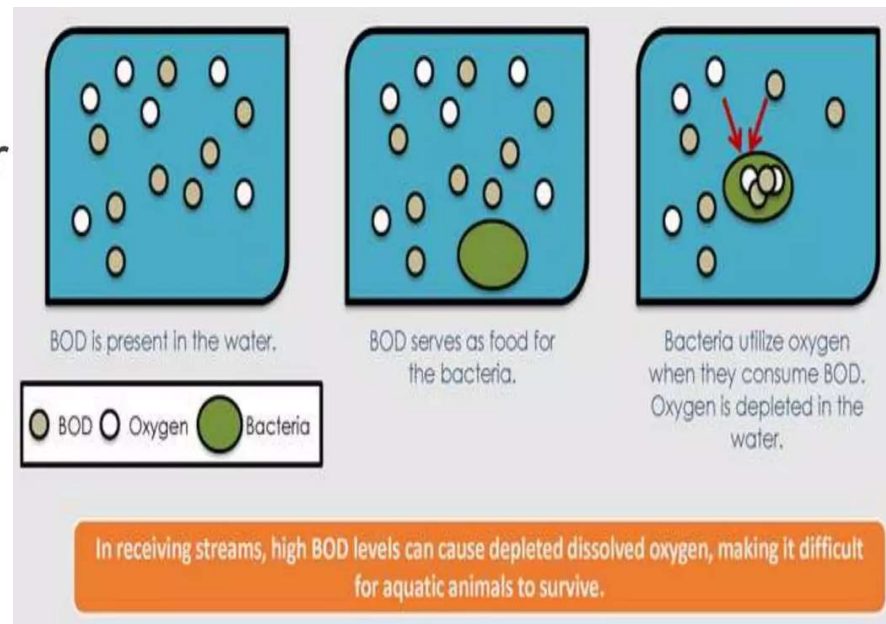
Biological Oxygen Demand (BOD, also called Biochemical Oxygen demand) is the amount of dissolved oxygen needed (i.e. demanded) by aerobic biological organisms to break down organic material present in a given water sample at certain temperature over specific time period .

Firstly remember few important points

1. Biochemical oxygen demand is a measure of the quantity of the oxygen used by microorganisms (e.g.. Aerobic bacteria) in the oxidation of the organic matter .
2. Most of the bacteria in the aquatic water column are aerobic . That means that they use oxygen to perform their metabolic activities of decomposition.
3. If dissolved oxygen concentration drop below 5 parts per million (ppm), fish will be unable to live for very long .

SIGNIFICANCE OF BOD

- ❖ To check quality of water
- ❖ Biologically oxidisable substance (e.g. carbohydrates) increases the BOD of water
- ❖ When BOD is high , dissolved oxygen becomes low
- ❖ Greater the BOD greater is the pollution .
- ❖ The demand of oxygen is proportional to the amount of organic waste to be degraded aerobically .



How BOD of Waste water is determined?

- ❖ The most common and accurate method is the 5-day BOD test.
 - ❖ This test involves incubating a sample of water for 5 days at 20 degrees Celsius.
 - ❖ The amount of **dissolved oxygen** in the water is then measured at the beginning and end of the 5 days.
 - ❖ The difference in these two measurements is the BOD5 value.
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- ❑ We can't determine the biowaste directly because it is already consumed by Microorganisms.
 - ❑ However by determining the DO consumed by MO, the biowaste can be determined.

DO consume by MO = Biologically oxidizable waste consumed by MO

Higher the DO consumed by MO, higher is the BOD of wastewater

Winklers Iodometric Method

CHEMICAL OXYGEN DEMAND (COD)

WHAT IS COD ?

Chemical oxygen demand is the amount of oxygen needed to oxidize the organic matter present in water. Chemical oxygen demand testing is used to determine the amount of oxidation that will occur and the amount of organic matter in a water sample. Chemical oxygen demand testing is also used to determine the amount of inorganic chemicals in a sample.

SIGNIFICANCE OF COD

- ❖ COD values are greater than BOD values .
- ❖ Pollution measuring factor
- ❖ Time period – 3 hours .
- ❖ COD values are not effected by toxins .
- ❖ Proposing standard for discharging domestic or industrial effluents.

Determination of COD

There are two methods available for COD determination namely open reflux and closed reflux.

Open Reflux Principle:

- Suitable for a wide range of wastes with a large sample size.
- Due to its higher oxidizing ability dichromate reflux method is preferred over other procedures using other oxidants (e.g. potassium permanganate).
- Oxidation of most organic compounds is up to 95-100% of the theoretical value.

Closed Reflux Principle:

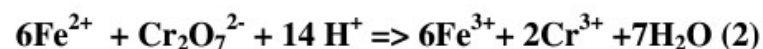
- This method is conducted with ampoules and culture tubes with pre-measured reagents.
- Measurement of sample volume and reagent volume are critical.
- This method is economical in the use of metallic salt reagents and generate smaller quantity of hazardous wastes.
- Volatile organic compounds (VOC) gets completely oxidized in a closed system than the open because of longer contact time with oxidants.

Chemical Reactions:



Here $d = (2n/3) + (a/6) - (b/3) - (c/2)$

During experiment, excess dichromate concentration is determined by titrating it with ferrous ammonium sulfate (FAS). The reaction is given by:



Here $d = (2n/3) + (a/6) - (b/3) - (c/2)$

COD is given by

$$\text{COD (mg O}_2\text{ /L)} = [(A-B) \times M \times 8000] / (V_{\text{sample}})$$

Where:	A	=	volume of FAS used for blank (mL)
	B	=	volume of FAS used for sample (mL)
	M	=	molarity of FAS
	8000	=	milli equivalent weight of oxygen (8) × 1000 mL/L.

E Waste

Electronic waste refers to **discarded electrical** or **electronic devices** that have reached the **end of their life cycle** or are no longer valuable to their current owners.

Fastest Growing Waste Stream:

E-waste is the fastest growing **solid waste stream** globally, increasing three times faster than the world's population.

In 2019, approximately **53.6 million tonnes** of e-waste were produced worldwide, but only **17.4%** of it was formally collected and recycled.

Common E-Waste Items:

Common items in e-waste streams include:

- ☐ Computers
- ☐ Mobile phones
- ☐ Large household appliances
- ☐ Medical equipment

Environmental and Health Impacts:

1. When e-waste is not treated, disposed of, and recycled appropriately, it can become a **threat to the environment and human health**.
2. Inferior recycling methods can release as many as **1000 different chemical substances** into the environment, including harmful neurotoxins like lead.
3. **Pregnant women and children** are particularly in danger due to their unique exposure pathways and developmental status.

Global Challenges:

- E-waste poses significant risks, especially in low- and middle-income countries (LMICs) where appropriate regulations, recycling infrastructure, and training are lacking.
- E-waste is considered **hazardous waste** because it contains **toxic materials** or can **produce toxic chemicals** when treated improperly.

Valuable Resources:

- E-waste contains valuable and finite resources that can be reused if recycled appropriately.
- It has become an important income stream for individuals and communities.

Prevention and Management:

- To address e-waste, we need better recycling practices, awareness, and regulations.
- Proper recycling facilities and training are essential to minimize health risks and environmental harm.

Green Chemistry

Definition and Principles:

- **Green chemistry** is the design of chemical products and processes that **reduce or eliminate** the use or generation of **hazardous substances**.
- It applies across the entire life cycle of a chemical product, including its **design, manufacture, use, and ultimate disposal**.

Applications:

- **Pharmaceuticals:** Developing drugs with reduced toxicity and more efficient synthesis methods.
- **Materials Science:** Creating eco-friendly polymers, coatings, and nanomaterials.
- **Energy:** Designing cleaner fuels, batteries, and solar cells.
- **Agrochemicals:** Pesticides and fertilizers with minimal environmental impact.
- **Industrial Processes:** Improving manufacturing processes to reduce waste and emissions.

12 Principles of Green Chemistry

1.Prevention:

1. It is better to **prevent waste** than to treat or clean it up after it has been created.
2. Focus on minimizing waste generation from the outset.

2.Atom Economy:

1. Synthetic methods should be designed to **maximize the incorporation** of all materials used in the process into the final product.
2. Aim for efficient utilization of reactants.

3.Less Hazardous Chemical Syntheses:

1. Design chemical reactions that use **safer reagents** and conditions.
2. Avoid toxic or hazardous substances whenever possible.

4.Designing Safer Chemicals:

1. Develop chemicals that have **minimal toxicity** to humans, animals, and the environment.
2. Prioritize safety during chemical design.

5.Safer Solvents and Auxiliaries:

1. Choose **benign solvents** and reaction conditions.
2. Reduce reliance on harmful solvents.

6.Design for Energy Efficiency:

1. Optimize processes to **minimize energy consumption**.
2. Use energy-efficient methods whenever feasible.

12 Principles of Green Chemistry

1. Use of Renewable Feedstocks:

1. Incorporate **renewable and sustainable raw materials** (feedstocks) into chemical processes.
2. Reduce dependence on fossil fuels.

2. Reduce Derivatives:

1. Minimize the use of unnecessary derivatives (intermediate compounds).
2. Simplify synthetic routes to reduce waste.

3. Catalysis:

1. Use **catalysts** to enhance reaction efficiency.
2. Catalysts promote specific reactions without being consumed.

4. Design for Degradation:

1. Create products that **degrade into non-hazardous substances** after use.
2. Facilitate environmentally friendly disposal.

5. Real-time Analysis for Pollution Prevention:

1. Implement monitoring and analysis during chemical processes.
2. Detect and prevent pollution in real time.

6. Inherently Safer Chemistry for Accident Prevention:

1. Design processes and products to be **inherently safer**.
2. Minimize risks to workers, the public, and the environment.